

SEBASTIAN TUDOR GHERGHE

Toronto, Ontario | +1 647-974-4844 | <https://gherghe.github.io/>
sebi96gherghe@gmail.com | [linkedin.com/in/sebastian](https://www.linkedin.com/in/sebastian) | github.com/sgherghe

EDUCATION

PhD in Mathematics

2019–2024

Department of Mathematics, University of Toronto. 3.96/4.00.

Toronto, ON

- Thesis: [Adiabatic quantum molecular dynamics](#), sup. Prof. Israel Michael Sigal.
- Awards: QEII/Halperin OGS (2023), Ida Bulat Teaching Award for Graduate CIs (2023).

MSc in Mathematics

2018–2019

Department of Mathematics, University of Toronto.

Toronto, ON

- Mathematical analysis of Schrödinger-Poisson systems, sup. Prof. Israel Michael Sigal.
- Awards: Margaret Isobel Elliott (2019).

B.Sc. (Hons) in Mathematics and Physics

2014–2018

University of Toronto.

Toronto, ON

- Awards: George Roderick Fraser (GRF) Admission (2014), GRF Scholar (2015–2017).

TECHNICAL SKILLS

Languages: Python (pandas, NumPy, scikit-learn, Matplotlib), L^AT_EX, Git.

Data & ML: data wrangling, linear regression, binary classification, dimensionality reduction (PCA), basic time-series analysis.

Math: PDEs, spectral theory, many-body quantum mechanics.

EXPERIENCE

Data Science Bootcamp

2021

The Erdős Institute. [Certificate](#).

Online

- Collaborated on a final project that involved extracting and analyzing data from almost 3 000 World Bank loan agreements to perform regional and textual analyses.
- Tools: Python (pandas, NumPy, scikit-learn, NLTK, Matplotlib), PCA, GitHub.

Course Instructor

2023–2024

Department of Mathematics, University of Toronto

Toronto, ON

- Lectured 7 undergraduate courses in mathematics (class sizes up to 150).
- Designed syllabi, exams, and managed teams of up to 3 teaching assistants.

PROJECTS

Olliver-Ricci curvature as a fragility indicator

2024

Department of Mathematics, University of Toronto

Toronto, ON

- Co-authored a [preprint](#) investigating an optimal transport graph-theoretic stock market fragility indicator. Applied it to S&P 500, NDXT, and TMX data; showed it quantifies crisis depth and duration (e.g. Lehman 2008).
- Tools: Python (pandas, NumPy, yfinance), optimal transport, graph theory, GitHub.

Academic research

2019–2024

Department of Mathematics, University of Toronto

Toronto, ON

- On the Time-Dependent Born-Oppenheimer Approximation (2024), *S. Gherghe, I. Moyano, and I. M. Sigal*. Submitted; [preprint](#).
- On the Mixed Quantum-Classical Approximation (2024), *S. Gherghe and I. M. Sigal*. Preprint.
- Global Spectral Gap in Bosonic Molecular Hamiltonians (2023), *S. Gherghe*. [Preprint](#).